where

$$A = \frac{R}{1-\gamma} \tag{23}$$

and $\gamma \equiv 1/\alpha$. This aggregation result is reminiscent of the classic aggregation result of Houthakker (1955–1956).[20] The factor A is what macroeconomists normally refer to as TFP. Its level depends on α, a parameter of the primitive distribution of productivity shocks, as well as on all the characteristics of the labour market as summarized by the destruction decision, R.

5. LABOUR-MARKET POLICIES AND THE LEVEL OF TFP

This section considers the effects of four policies: employment subsidies, hiring subsidies, firing taxes, and unemployment benefits. I follow Pissarides (2000) and model the subsidies as transfers from the government to the firm and the firing taxes as payments from the firm to the government.[21] The value function $W(x)$ is still given by (5), while (4), (6), and (7) generalize to

$$rU = \tau_b + \theta q(\theta) \int \max[W_o(z) - U, 0] dG(z)$$

$$rV = -ck + q(\theta) \int \max[J_o(z) + \tau_h k - V, 0] dG(z)$$

$$rJ(x) = \pi(x) + \tau_e k + \lambda \int \max[J(z) - V + \tau_f k, 0] dG(z) - (\delta + \lambda)[J(x) - V + \tau_f k].$$

The policy variables are a hiring subsidy, τ_h, an employment subsidy, τ_e, a firing tax, τ_f, and unemployment benefits, τ_b.[22] In Appendix B I show that if G is as in (20), the job-destruction and job-creation conditions that determine the equilibrium pair (θ, R) are

$$R - \phi - c + \tau_e + r\tau_f - \left(\tau_b + \frac{\beta}{1-\beta} c\theta\right) + \frac{\lambda \varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} = 0 \tag{24}$$

$$\frac{\varepsilon^\alpha R^{1-\alpha}}{(\alpha - 1)(r + \delta + \lambda)} + \left(\frac{\varepsilon}{R}\right)^\alpha (\tau_h - \tau_f) - \frac{c}{(1-\beta) q(\theta)} = 0. \tag{25}$$

The main properties of the equilibrium are summarized as follows.

Proposition 1. *Suppose* $\varepsilon + (\alpha - 1)(r + \delta + \lambda)(\tau_h - \tau_f) > 0$. *Let* θ_ϵ^* *be defined by* $q(\theta_\epsilon^*) = \frac{(\alpha-1)(r+\delta+\lambda)c}{(1-\beta)[\varepsilon+(\alpha-1)(r+\delta+\lambda)(\tau_h-\tau_f)]}$, *and* $\phi_\epsilon = \left[1 + \frac{\lambda}{(\alpha-1)(r+\delta+\lambda)}\right]\varepsilon - \left(1 + \frac{\beta}{1-\beta}\theta_\epsilon^*\right)c + \tau_h + r\tau_f - \tau_b$. *If* $\varepsilon + \alpha(r + \delta + \lambda)(\tau_h - \tau_f) > 0$, *then for any* $\phi > \phi_\varepsilon$ (a) *there exists a unique equilibrium,* (b) $R > \varepsilon$, (c) $\partial R/\partial\phi > 0$, *and* (d) $\partial\theta/\partial\phi < 0$. *If in addition* $\phi_\epsilon - \varepsilon > 0$, *then* (e) *there is a nondegenerate interval* $(\phi_\epsilon, \widetilde{\phi})$ *such that* $R(\phi) < \phi$ *for all* $\phi \in (\phi_\epsilon, \widetilde{\phi})$.

Aggregate output is still given by (22); the aggregate stock of capital in filled jobs, K_e, is still given by (17); and the aggregate number of hours worked, N, is still given by $N = (R/\mu)^\alpha K_e$. In addition, if the measure of capital used to construct aggregate output is K_e, then the level of TFP

20. I discuss this relationship at some length in Section 6.

21. I assume that upon separation, the firm must pay the firing tax to the government because in the present set-up, firing taxes would be completely neutral under the alternative scheme in which the firm compensates the fired worker directly.

22. All payments are assumed to be proportional to the firm's size, as measured by k. For example, upon separation, the firm is taxed $\tau_f k$, and while unemployed, the worker receives $\tau_b k$ from the government. This assumption, borrowed from Pissarides (2000), is convenient because it ensures that policies introduce no scale effects into the job-creation and job-destruction decisions. Also, to keep the analysis simple, I will ignore the government's financing constraints. A natural extension would be requiring the government to run a balanced budget. A simple example of a scheme that is self-financing in the steady state is $\tau_f = \tau_h$, and $\tau_b = \tau_e = 0$.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms

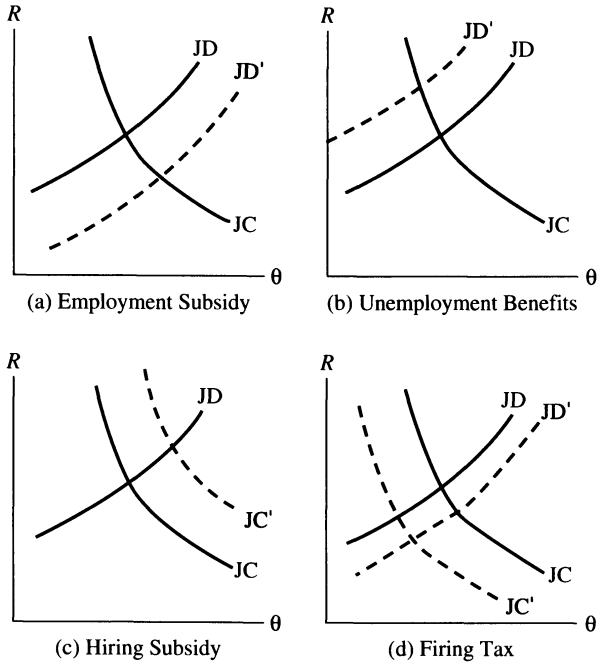

FIGURE 1

Equilibrium effects of policies on job-creation (JC) and job-destruction (JD) rates

is still as in (23). The following result, which holds under the assumptions stated in Proposition 1, summarizes the effects that labour-market policies have on the level of TFP.

Proposition 2. *Employment subsidies and firing taxes reduce A. Hiring subsidies and unemployment benefits increase A.*

Since A is proportional to R, policies have the same qualitative effect on TFP as they have on the destruction rate. Proposition 2 is illustrated in Figure 1. In each panel, the dashed lines labelled JC′ and JD′ indicate how the curve shifts in response to the introduction of the policy.

Employment subsidies make firms more tolerant of low productivity realizations and hence lower the average productivity of active firms. All else equal, an economy with relatively high subsidies to continued employment will exhibit a low job-destruction rate, a high job-creation rate, and hence low levels of unemployment and measured TFP. Firing taxes have a similar qualitative effect on job destruction, but that mechanism is reinforced by a relatively low rate of job creation (which reduces the reservation wage and hence makes firms even more tolerant of low productivity realizations). So firing restrictions will reduce measured TFP, as well as the job-creation and job-destruction rates. Hiring subsidies have no direct effect on the destruction decision, but they do stimulate job creation. This increases market tightness, which in turn increases the worker's outside option and raises measured TFP, job creation, and job destruction. Unemployment benefits also cause R to rise through an increase in the worker's reservation wage. Consequently, economies with relatively high unemployment benefits will tend to exhibit relatively high levels of TFP and unemployment.[23]

23. Notice that these results obtain if factor inputs are measured rather accurately, so that the TFP residual corresponds to (23). But naturally, what is measured as the TFP residual, as well as how it responds to policy, *will* depend on the accuracy with which factor inputs are measured. See Sections 6.2 and 7 below.

© 2006 Federal Reserve Bank of Minneapolis

This content downloaded from
158.143.193.212 on Fri, 19 Nov 2021 13:09:47 UTC
All use subject to https://about.jstor.org/terms